Briefings in Bioinformatics, 2026, pp. 1–17

doi:
Paper

PAPER

G2DR: A Genotype-First Framework for Genetics-Informed Target Prioritization and Drug Repurposing

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Abstract

Human genetics offers a promising route to therapeutic discovery, yet practical frameworks that translate genotype-derived signal into ranked computational hypotheses for candidate targets and repurposing candidates remain limited, particularly in settings where matched disease transcriptomics are unavailable. Here we present G2DR, a genotype-first computational prioritization framework that propagates inherited genetic variation through genetically predicted gene expression, multi-method gene-level testing, pathway enrichment, network context, and multi-source drug–target evidence integration to generate ranked hypotheses for candidate targets and associated compounds for downstream experimental follow-up. In a migraine case study using 733 UK Biobank participants (53 cases and 680 controls) under stratified five-fold cross-validation, we imputed genetically regulated expression across multiple transcriptome prediction resources and ranked genes using a reproducibility-aware composite score derived exclusively from training and validation data. Prioritized genes were evaluated on a held-out test split. In internal held-out evaluation, discovery-based prioritization generalized to unseen data, achieving a gene-level ROC-AUC of 0.775 and PR-AUC of 0.475 for recovery of test-significant genes under the defined ranking framework. The integrated ranking showed enrichment for curated migraine-associated biology while retaining substantial novel candidate space. Mapping prioritized genes to compounds through Open Targets, DGIdb, and ChEMBL produced candidate drug sets enriched for migraine-related and literature-associated compounds relative to a global drug background. G2DR is intended as a modular computational prioritization engine for hypothesis generation and follow-up prioritization in genotype-first settings, not as a clinically actionable target-identification system. All outputs require independent experimental and clinical validation. Supplementary data are available at *Briefings in Bioinformatics* online.

Key words: drug repurposing, human genetics, TWAS, genetically predicted expression, target prioritization, translational bioinformatics

Introduction

Drug repurposing remains an attractive strategy for accelerating therapeutic development because existing compounds benefit from prior pharmacological, toxicological, and manufacturing knowledge ([1, 2, 3]). Over the past two decades, computational repurposing has expanded from expression-signature matching ([4]) and network medicine ([5, 6, 7]) to machine-learning and knowledge-graph approaches that integrate heterogeneous biomedical evidence at scale ([8, 9]). Yet many current pipelines depend on disease-specific transcriptomic profiles ([10, 11]), curated disease modules ([6, 12]), or historical drug–disease labels ([2, 8]), which can be limiting for newly studied phenotypes or for biobank settings where genotype and phenotype labels are available but disease-relevant molecular profiling is sparse or absent ([13, 14, 2, 15]).

Human genetics provides a particularly attractive starting point for therapeutic prioritization because inherited variation is stable, scalable, and increasingly well characterized across large cohorts ([13, 14]). When measured expression is unavailable, genotype can be propagated through transcriptome imputation models to estimate genetically regulated expression ([16, 17]), offering an interpretable intermediate layer between variant-level signal and candidate genes ([15]). This strategy is appealing, but it also comes with important caveats: TWAS-style approaches prioritize genes associated with trait-linked regulatory signal, not necessarily causal effector genes ([15]), and inference remains sensitive to linkage disequilibrium, co-regulation, tissue context, ancestry, and model architecture ([15, 18]). These limitations argue for integrative frameworks that improve robustness and interpretability rather than relying

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Table 1. Expression weight models used for genotype-based gene expression prediction.

Model	Source	Description	Gene–Tissue Pairs
MASHR	`https://zenodo.org/records/7551845`	Multivariate adaptive shrinkage models for cross-population transcriptome prediction, designed to improve TWAS performance in underrepresented populations [26]	686,241 (49 tissues)
JTI	`https://zenodo.org/records/3842289`	Joint-tissue imputation models that borrow information across tissues via shared genetic regulation, enabling both expression prediction and downstream causal inference [27]	533,141 (49 tissues)
CTIMP	`https://github.com/yiminghu/CTIMP`	Cross-tissue gene-expression imputation models trained with sparse group-LASSO to borrow information across GTEx tissues [28]	404,450 (49 tissues)
UTMOST	`https://zenodo.org/records/3842289`	Cross-tissue transcriptome prediction models from the UTMOST framework, hosted alongside JTI models; treated as a separate model set in this study [28]	340,104 (49 tissues)
EpiXcan	`https://www.synapse.org/Synapse:syn52745052`	Transcriptome imputation incorporating epigenomic annotations to weight SNP priors, improving prediction accuracy over standard elastic net [29]	143,609 (14 tissues)
FUSION	`http://gusevlab.org/projects/fusion/`	Bayesian sparse linear mixed model and penalized regression framework for transcriptome-wide association studies [17]	265,052 (48 tissues)
TIGAR	`https://github.com/yanglab-emory/TIGAR`	Nonparametric Bayesian gene expression imputation tool supporting individual- and summary-level GWAS data for transcriptome-wide association studies [30]	181,605 (49 tissues)

on any single transcriptome-prediction resource or analytical test ([15]).

A further motivation comes from therapeutic translation. Multiple studies have shown that targets supported by human genetics are more likely to succeed in clinical development ([19, 20]), making genetics-informed target selection an increasingly important component of drug discovery. However, the key translational challenge is not simply to detect associated genes, but to prioritize those with the strongest convergent support and then connect them to tractable pharmacological hypotheses in a transparent and reproducible manner.

Here, we present G2DR, a genotype-first computational framework for genetics-informed target prioritization and drug repurposing hypothesis generation. G2DR integrates genetically predicted expression from multiple transcriptome-weight resources ([16, 17, 15]), multi-method gene-level testing, pathway enrichment, network context, structure- and knowledge-based druggability, and multi-source drug–target evidence ([21, 22, 23, 24, 25]) to generate ranked computational hypotheses for candidate targets and associated compounds, without requiring measured disease transcriptomics or supervised training on curated indication labels ([2]). We demonstrate the framework in migraine using UK Biobank data and evaluate performance at multiple levels, including held-out gene replication, enrichment for curated migraine biology, and recovery of migraine-linked drugs from prioritized targets. Framed explicitly as a computational prioritization engine for downstream experimental follow-up rather than a clinically actionable target-identification system, G2DR is best viewed as a modular proof-of-concept framework with clear room for further maturation.

Methods

Study Design and Cohort Assembly

A total of 733 participants (53 cases and 680 controls) who reported migraine and recorded comorbid conditions, including hypertension, asthma, depression, osteoarthritis, hypercholesterolemia, irritable bowel syndrome, hypothyroidism, hay fever, and gastroesophageal reflux disease, were extracted from the UK Biobank ([13, 14]). These comorbidities are commonly reported to co-occur with migraine ([31, 32, 33, 34]). Genotype–phenotype data were partitioned using stratified five-fold cross-validation, with each fold split into training (80%), validation (10%), and test (10%) subsets to preserve the case–control ratio. Genotype quality control was applied within each training fold and propagated to validation and test subsets, including per-variant filters for minor allele frequency (MAF) ≥ 0.01, Hardy–Weinberg equilibrium (HWE) $p \geq 1 \times 10^{-6}$, and variant missingness ≤ 0.1, along with per-individual missingness ≤ 0.1 and removal of related individuals using PLINK (`--rel-cutoff 0.125`) ([35, 36]).

Genetically Predicted Gene Expression

Genetically predicted gene expression was computed for each individual within each fold using PrediXcan-style genotype-based transcriptome imputation ([16, 18]), in which expression for gene g in tissue t is estimated as a weighted sum of SNP dosages using pre-trained expression-weight models. To maximize gene coverage and capture diverse regulatory architectures, we incorporated seven publicly available expression-weight databases trained under different statistical and biological assumptions: MASHR, JTI, CTIMP, UTMOST, EpiXcan, FUSION, and TIGAR (Table 1; full mathematical details are provided in Supplementary Methods S1). Predicted expression values were adjusted for sex and the top 10 genetic principal components using ordinary least squares regression applied to the training split only, with fitted parameters applied to validation and test splits.

Differential Expression Analysis

Differential expression analysis (DEA) was performed on covariate-adjusted imputed expression values for each gene, tissue, database, fold, and dataset split using eight statistical methods: LIMMA empirical Bayes moderated t-test ([37]), Welch's unequal-variance t-test, OLS regression, Wilcoxon rank-sum test, phenotype-label permutation test ($B = 1{,}000$ permutations), weighted logistic regression, bias-reduced Firth logistic regression, and Bayesian logistic approximation. For differential-expression methods, significance required FDR < 0.1 and $|\log_2 \mathrm{FC}| \geq 0.5$; for association-style methods, significance required FDR < 0.1 and $|\mathrm{Effect}| \geq 0.5$. Full model specifications are provided in Supplementary Methods S1.

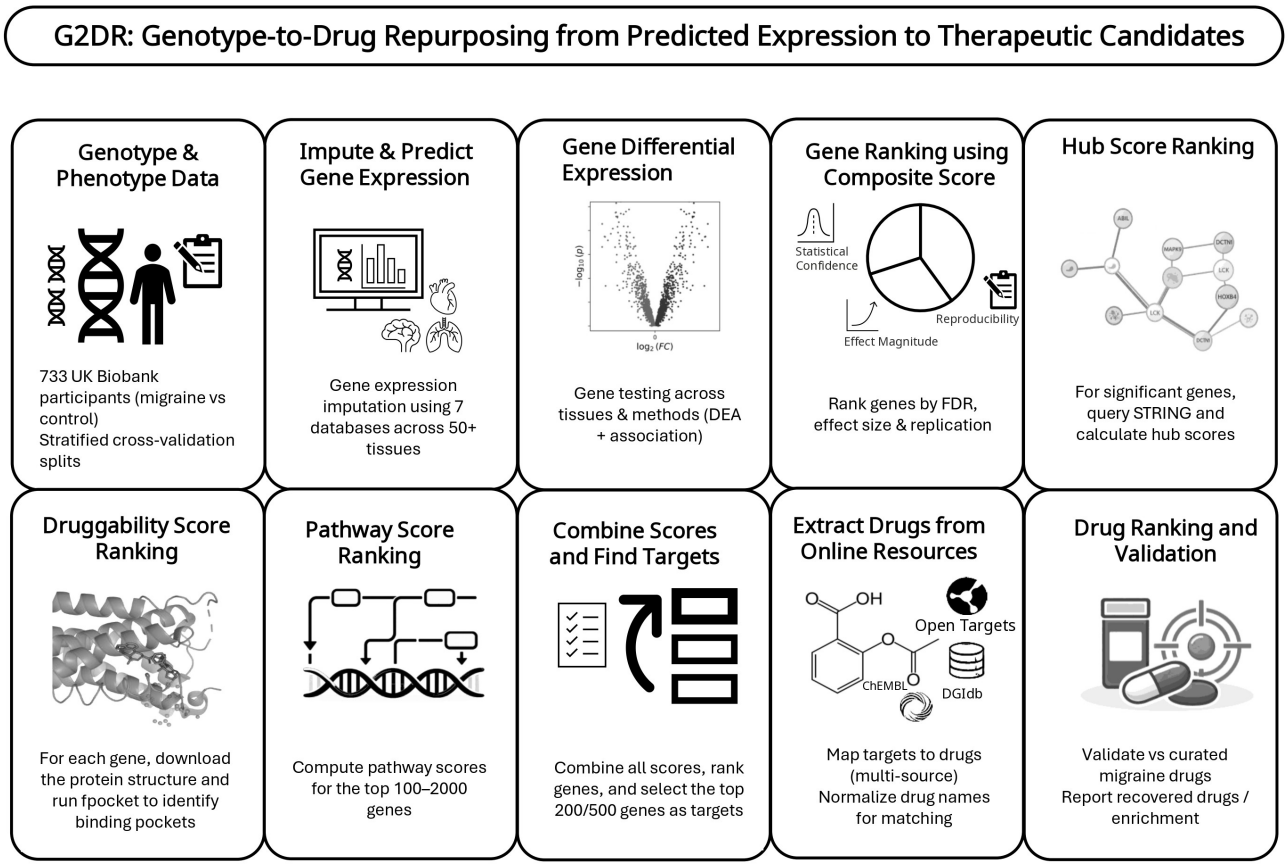

Fig. 1. Overview of the G2DR framework. Genotype and phenotype data from the migraine cohort were partitioned by stratified cross-validation and propagated through genotype-based transcriptome imputation across multiple expression-weight resources. Covariate-adjusted predicted expression values were tested using multiple differential-expression and association methods, and significant results from the training and validation splits were aggregated into a discovery set. Genes were then ranked using a composite score that integrated reproducibility, effect magnitude, and statistical confidence, followed by pathway, network, and druggability annotation to generate an integrated target-prioritization score. Top-ranked genes were mapped to candidate compounds through Open Targets, DGIdb, and ChEMBL, and the resulting drug lists were evaluated against curated migraine-associated drug references.

Gene Prioritization

Significant findings were aggregated across all databases, tissues, folds, and methods. The discovery set was defined as $G_{\text{discovery}} = G_{\text{train}} \cup G_{\text{val}}$, reserving G_{test} exclusively for out-of-sample evaluation. Each gene in $G_{\text{discovery}}$ was scored using a composite importance score S_g that integrates three components: reproducibility (40%), effect magnitude (30%), and statistical confidence (30%). Reproducibility captures both the total number of significant hits and the breadth of support across database–tissue–method combinations. Effect magnitude is derived from method-standardized absolute effect sizes. Statistical confidence is derived from BH-adjusted FDR values. Full scoring formulas are provided in Supplementary Methods S2. Rankings were robust to alternative weights (Spearman $\rho = 0.992$; Top-100 overlap $\geq 75\%$; full stability results in Supplementary Table S1).

Generalizability was assessed using the held-out test split as the replication target. For each gene g in the tested universe U, a binary label $y_g = \mathbb{I}(g \in G_{\text{test}})$ was defined, and S_g was used as the ranking predictor. Performance was quantified using ROC-AUC, PR-AUC, and hypergeometric enrichment tests at multiple top-K cutoffs. Enrichment for a curated set of 3,190 migraine-associated genes ([15]) assembled via `PhenotypeToGeneDownloaderR` (`https://github.com/MuhammadMuneeb007/PhenotypeToGeneDownloaderR`) was assessed using the same hypergeometric framework. Full test derivations are provided in Supplementary Methods S3.

To obtain an integrated target-prioritization ranking, S_g was combined with three additional evidence layers: pathway support (*PathwayScore*) derived from GO, KEGG, Reactome, and Disease Ontology enrichment analyses (`clusterProfiler`, `ReactomePA`, `DOSE`); network hub score from STRING protein–protein interaction analysis; and structure- or knowledge-based druggability from DGIdb, ChEMBL, and fpocket. The integrated score was computed as $\text{CoreScore}_g = 0.45 \cdot \text{DE}_g^{\text{norm}} + 0.25 \cdot \text{Path}_g^{\text{norm}} + 0.25 \cdot \text{Drug}_g^{\text{norm}} + 0.05 \cdot \text{Hub}_g^{\text{norm}}$, with DE weighted highest because it provides the primary TWAS-derived genetic link between genotype and disease. Full pathway score propagation formulas and weight justification are provided in Supplementary Methods S2.

Drug Repurposing

A curated migraine drug reference set of 4,824 normalized drug entries was assembled using `DownloadDrugsRelatedToDiseases` (`https://github.com/MuhammadMuneeb007/DownloadDrugsRelatedToDiseases`), which aggregates drug–disease associations

from Open Targets, DrugBank, CTD, and literature-mining sources. All enrichment claims are framed relative to a global background of 139,597 drugs. Candidate compounds were mapped from the top N ranked genes using drug–target evidence from Open Targets, DGIdb, and ChEMBL, with drug names harmonized using conservative text normalization. Each gene–drug evidence row was assigned an evidence weight based on clinical development stage, and a DrugScore was computed by aggregating GeneWeight × EvidenceWeight contributions across all supporting genes.

The curated migraine drug reference set was partitioned into four development-stage tiers: Tier 1 (migraine-specific approved therapies), Tier 2 (guideline-supported acute or preventive therapies), Tier 3 (established off-label therapies), and Tier 4 (broader literature-linked compounds). Performance was assessed using multi-K overlap evaluation under two complementary universes: an ALL-drugs global background (hypergeometric test valid) and a PREDICTED candidates-only universe (AUROC and AUPRC are the appropriate metrics; full results in Supplementary Table S5).

Directionality Assessment

To assess mechanistic compatibility of prioritized gene–drug pairs, drug action annotations were extracted from locally aggregated evidence fields and supplemented via ChEMBL and DGIdb, harmonized into a reduced action vocabulary (inhibitor, antagonist, agonist, activator, modulator, unknown). Each pair was classified as directionally consistent when the drug action was compatible with the inferred disease-associated gene direction, inconsistent when opposed, or unclear when insufficient annotation was available. Full annotation and classification criteria are provided in Supplementary Methods S4.

Results

Cross-database concordance of predicted gene expression

A total of 733 participants (53 cases, 680 controls) were stratified into five cross-validation folds, preserving case–control balance across training (80%), validation (10%), and test (10%) splits. Within each fold and split, we imputed genetically regulated expression from genotype dosages using PrediXcan-style models ([16, 18]).

To compare expression-weight databases, we quantified tissue-matched concordance by computing gene-wise Pearson correlations of predicted expression across individuals for genes shared by each database pair, then averaging within tissue and across folds to yield one concordance estimate per database pair and split. The highest concordance was observed for the CTIMP–UTMOST pair ($r \approx 0.55$; train = 0.547, validation = 0.546, test = 0.546), followed by JTI–CTIMP ($r \approx 0.53$; train = 0.526, validation = 0.525, test = 0.525), MASHR–UTMOST ($r \approx 0.54$; train = 0.536, validation = 0.538, test = 0.537), and MASHR–CTIMP ($r \approx 0.49$; train = 0.491, validation = 0.492, test = 0.492). Concordance with FUSION was moderate for UTMOST ($r \approx 0.50$; train = 0.498, validation = 0.497, test = 0.499) and lower for MASHR ($r \approx 0.38$; train = 0.380, validation = 0.380, test = 0.378) and JTI ($r \approx 0.31$; train = 0.311, validation = 0.310, test = 0.311). TIGAR showed near-zero concordance with all other databases ($r \approx 0.03$), indicating substantially different predicted-expression patterns (Figure 2).

Adjusting expression for sex and the top ten genetic principal components had negligible impact on cross-database correlations (overall mean $\Delta r = r_{\text{fixed}} - r_{\text{raw}} \approx -1 \times 10^{-4}$). However, the adjustment altered expression values modestly overall ($\text{corr}(\text{raw}, \text{fixed}) = 0.9696$, $\text{mean}|\Delta| = 0.00797$, RMSE = 0.0353), with the largest shifts observed in FUSION ($\text{mean}|\Delta| \approx 0.023$, RMSE ≈ 0.124) and in EpiXcan ($\text{mean}|\Delta| \approx 0.011$, RMSE ≈ 0.078), while TIGAR showed minimal absolute change ($\text{mean}|\Delta| \approx 6.3 \times 10^{-5}$). These results suggest that cross-database differences primarily reflect methodological and architectural variation among expression-weight resources, rather than confounding by sex or ancestry principal components.

Gene differential expression analysis

We performed 172,868,680 gene-level differential expression and association tests across expression-weight databases, five cross-validation folds, train/validation/test splits, and tissues using eight analytical methods (five differential expression and three association approaches). Applying method-specific thresholds (FDR < 0.1, $|\log_2 \text{FC}| \geq 0.5$ for differential-expression methods, and $|\text{Effect}| \geq 0.5$ for association-style methods) yielded 96,502 significant test records, corresponding to 11,451 unique significant genes from an overall tested universe of $U = 34{,}355$ genes. Split-wise, we detected 1,046 significant genes in training, 9,107 in validation, and 7,141 in the held-out test set. he comparatively higher count in validation and test splits likely reflects the modest cohort size: with only ∼5–6 cases per fold in validation and test, small-sample calibration can yield unstable significance calls. Importantly, the sensitivity analyses in Supplementary Table S2 confirm that the main prioritization conclusions—held-out ROC-AUC, fold-enrichment, and migraine-gene enrichment—are preserved under stricter thresholds (FDR < 0.05, $|\log_2 \text{FC}| \geq 0.75$), arguing against threshold permissiveness as the primary driver of the observed signal. Discovery candidates were defined without test leakage using training and validation only, producing $G_{\text{discovery}} = G_{\text{train}} \cup G_{\text{val}}$ ($n = 9{,}305$). We then prioritized genes in $G_{\text{discovery}}$ using a composite score integrating reproducibility, effect magnitude, and statistical confidence; annotation against a curated set of 3,190 known migraine genes indicated that $G_{\text{discovery}}$ contained 1,189 known genes and 8,116 novel candidates.

Out-of-sample replication was assessed by treating test-significant genes ($T = 7{,}141$) as positives within U, where $G_{\text{discovery}}$ recovered 4,995 test positives (recall = 0.699) with precision = 0.537, while smaller top-K lists achieved higher precision at the cost of recall (e.g., top-100 precision = 0.790; Table 2).

To quantify how well the continuous composite score prioritized genes that replicated in the held-out test split, we evaluated it as a genome-wide ranking model over the full tested universe U. Each gene was assigned a binary replication label, with positives defined as genes significant in the test set ($T = 7{,}141$) and negatives defined as all remaining genes in $U \setminus T$. The composite score achieved ROC-AUC = 0.7753 and PR-AUC = 0.4754, indicating meaningful separation of test-replicating genes from non-replicating genes in this full-universe ranking frame, where non-discovery genes were assigned a score of zero. Given the baseline prevalence of positives ($T/U = 0.2079$), the observed PR-AUC corresponds to a 2.29× lift

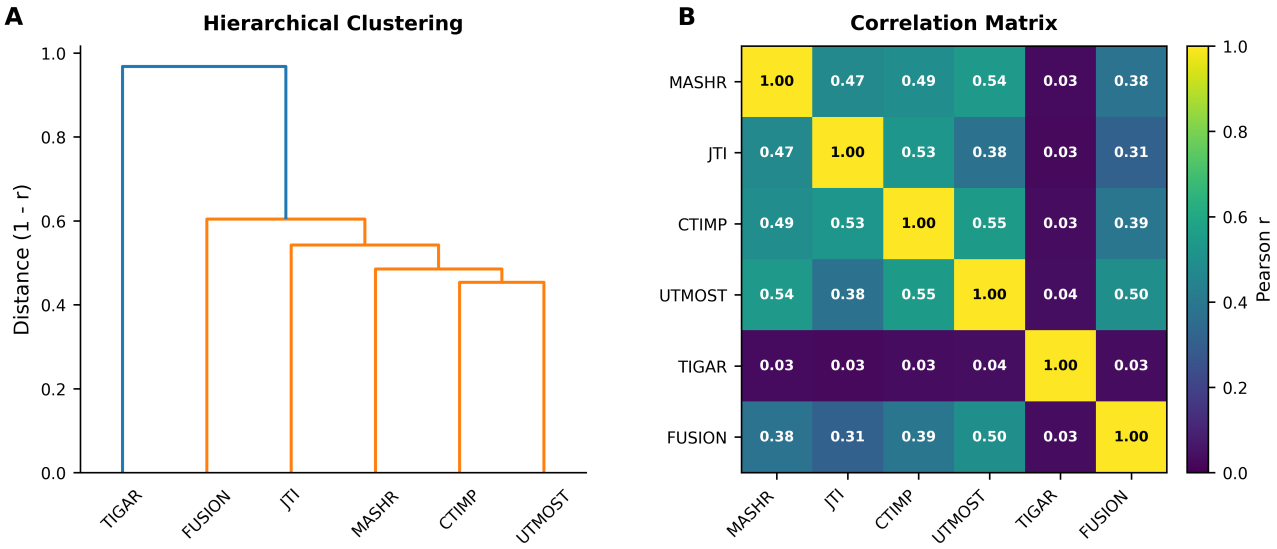

Fig. 2. Cross-database concordance of predicted gene expression across expression-weight databases. (A) Hierarchical clustering of databases using distance $(1 - r)$ derived from the pairwise correlation matrix. **(B)** Pairwise correlation matrix (Pearson r).

Table 2. Held-out test replication for the discovery set and top-K lists. Positives are test-significant genes ($T = 7{,}141$). TP/FP/FN are computed relative to T.

Predicted set	TP	FP	FN	Precision	Recall
$G_{\text{discovery}}$ (train ∪ val)	4,995	4,310	2,146	0.537	0.699
Top 50	39	11	7,102	0.780	0.005
Top 100	79	21	7,062	0.790	0.011
Top 200	156	44	6,985	0.780	0.022
Top 500	347	153	6,794	0.694	0.049
Top 1000	664	336	6,477	0.664	0.093

Table 3. Hypergeometric enrichment of predicted sets for held-out test positives and curated migraine genes. For test positives: $k_{\text{exp}} = N_{\text{pred}} \cdot (T/U)$, $T = 7{,}141$, $U = 34{,}355$. For migraine genes: $k_{\text{exp}} = N_{\text{pred}} \cdot (|M|/U)$, $|M \cap U| = 3{,}190$. FE $= k_{\text{obs}}/k_{\text{exp}}$.

	Test positives ($T = 7{,}141$)					**Curated migraine genes ($\lvert M\rvert = 3{,}190$)**				
Set	N_{pred}	k_{obs}	k_{exp}	**FE**	***p*-value**	N_{pred}	k_{obs}	k_{exp}	**FE**	***p*-value**
$G_{\text{discovery}}$ (train ∪ val)	9,305	4,995	1,934.13	2.58	$< 10^{-300}$	9,305	1,189	864.01	1.38	5.47×10^{-40}
Top 50	50	39	10.39	3.75	7.14×10^{-18}	50	12	4.64	2.58	1.71×10^{-3}
Top 100	100	79	20.79	3.80	1.55×10^{-35}	100	21	9.29	2.26	2.97×10^{-4}
Top 200	200	156	41.57	3.75	1.77×10^{-67}	200	36	18.57	1.94	8.72×10^{-5}
Top 500	500	347	103.93	3.34	6.94×10^{-123}	500	74	46.43	1.59	4.26×10^{-5}
Top 1000	1,000	664	207.86	3.19	2.30×10^{-220}	1,000	132	92.85	1.42	2.39×10^{-5}

over random ranking in precision–recall space. We next tested whether discovery sets were enriched for held-out test positives using a hypergeometric model over U. $G_{\text{discovery}}$ showed strong enrichment (2.58-fold; $p < 10^{-300}$). A ranking sweep from $K = 10$ to 2,000 identified maximal enrichment at $K = 10$ (4.33-fold), with enrichment remaining > 2.9-fold even at $K = 2{,}000$ (Table 3, left panel).

As an orthogonal validation, we tested enrichment for curated migraine genes ($M = 3{,}190$) within U. $G_{\text{discovery}}$ contained 1,189 known genes (expected 864.01), corresponding to 1.38-fold enrichment ($p = 5.47 \times 10^{-40}$), indicating that the discovery set preferentially captures established migraine biology while retaining a large novel component (Table 3, right panel).

Together, these results show that integrating gene-level evidence across tissues, methods, and databases yields a reproducible prioritization that generalizes to held-out test data, performs substantially better than random ranking, and can be further refined into compact candidate gene sets for downstream network and drug-mapping analyses.

Gene prioritization was robust to stricter significance thresholds

To assess whether the discovery-based prioritization was sensitive to the choice of significance threshold, we repeated the same workflow under stricter FDR and effect-size criteria. Relative to the primary analysis (FDR < 0.10, $|\log_2 \text{FC}| \geq 0.50$), tightening the FDR threshold to < 0.05 reduced the

number of significant genes in training, validation, and test from 1,046, 9,107, and 7,141 to 958, 7,861, and 5,313, respectively, and reduced $G_{\text{discovery}}$ from 9,305 to 8,135 genes. Increasing the effect-size threshold to $|\log_2 \text{FC}| \geq 0.75$ at FDR < 0.10 reduced these counts further to 263, 6,241, and 4,599 genes, yielding $G_{\text{discovery}} = 6{,}303$, while the strictest setting (FDR < 0.05, $|\log_2 \text{FC}| \geq 0.75$) produced 248 training genes, 5,709 validation genes, and 3,772 test genes, with $G_{\text{discovery}} = 5{,}794$.

Despite the expected reduction in $G_{\text{discovery}}$ size under stricter filtering, held-out replication remained substantial across all threshold settings. The primary analysis achieved ROC-AUC = 0.7753 and PR-AUC = 0.4754. Under FDR < 0.05 and $|\log_2 \text{FC}| \geq 0.50$, performance remained above random with ROC-AUC = 0.7314 and PR-AUC = 0.3412. Under the stricter effect-size threshold of $|\log_2 \text{FC}| \geq 0.75$, performance was similarly retained, with ROC-AUC = 0.7339 and PR-AUC = 0.3289 for FDR < 0.10, and ROC-AUC = 0.7025 and PR-AUC = 0.2497 for FDR < 0.05. $G_{\text{discovery}}$ also remained strongly enriched for held-out test positives across all settings, with fold enrichment ranging from 2.58-fold to 3.20-fold. Enrichment for curated migraine genes was likewise preserved, ranging from 1.38-fold under the primary rule to 1.41-fold under the strictest setting. Full sensitivity results are provided in Supplementary Table S2; together, these results indicate that the main gene-prioritization findings are not driven by permissive significance filtering.

Component-based recovery

To identify which components contributed the most disease-relevant signal in $G_{\text{discovery}}$, we tested whether each component-specific discovery set (defined using training and validation only) was enriched for curated migraine-associated genes. For each database, tissue, and method, we defined a discovery set as the unique genes that were significant at least once within that component across folds and the remaining dimensions, and quantified over-representation of known migraine genes using a fold-enrichment statistic (FE $= k_{\text{obs}}/k_{\text{exp}}$) with empirical p-values from size-matched random gene-set sampling. Full results across all databases, tissues, and methods are provided in Supplementary Table S3.

Across expression-weight databases, MASHR showed the strongest discovery enrichment (FE = 2.30, $p_{\text{emp}} = 1 \times 10^{-4}$), followed by JTI (FE = 1.86, $p_{\text{emp}} = 6.799 \times 10^{-3}$) and FUSION (FE = 1.36, $p_{\text{emp}} = 1 \times 10^{-4}$), whereas EpiXcan, CTIMP, UTMOST, and TIGAR showed weaker and non-significant enrichment. Across tissues, discovery enrichment highlighted a reproducible set of migraine-informative contexts, led by Brain Amygdala (FE = 1.96, $p_{\text{emp}} = 1 \times 10^{-4}$), Minor Salivary Gland (FE = 1.94, $p_{\text{emp}} = 1 \times 10^{-4}$), and Whole Blood (FE = 1.83, $p_{\text{emp}} = 1 \times 10^{-4}$), with multiple additional tissues showing significant over-representation of known migraine genes. Across methods, the discovery signal was dominated by the association-style models: Weighted Logistic and Bayesian Logistic both showed consistent enrichment (FE ≈ 1.37, $p_{\text{emp}} = 1 \times 10^{-4}$), while Welch t-test identified a smaller, disease-dense set (FE = 1.35, $p_{\text{emp}} = 6 \times 10^{-4}$); low-coverage methods did not show significant discovery enrichment.

Unified gene prioritization framework

We next evaluated a unified gene-prioritization framework by comparing individual evidence components against both the primary composite score and the multi-source integrated score. Two complementary evaluation universes were used throughout. The **full universe** comprised all $U = 34{,}355$ tested genes; genes not in the discovery set received a score of zero, so this universe measures how well a ranking separates discovery-relevant genes from the entire tested space — the same universe used to compute the headline ROC-AUC and PR-AUC reported above. The **discovery universe** comprised only the $n = 9{,}305$ genes in $G_{\text{discovery}}$ (train $\cup$ val); here non-discovery genes are absent, so this universe measures ranking quality *within* the already-selected candidates — a strictly harder question.

Two composite scores were evaluated. The **primary composite score** (S_g, Importance_Score) integrates reproducibility, effect magnitude, and statistical confidence with weights 40/30/30, and is the source of the headline test-replication metrics reported in the discovery section. The **integrated score** additionally combines S_g with pathway support, druggability evidence, and network hub score (weights DE = 0.45, Pathway = 0.25, Drug = 0.25, Hub = 0.05), and is designed for downstream target selection rather than pure statistical replication.

Evaluated over all 34,355 tested genes, effect-only ranking achieved the strongest test-replication performance (ROC-AUC = 0.790, PR-AUC = 0.526), marginally exceeding the primary composite (ROC-AUC = 0.775, PR-AUC = 0.475) and the integrated score (ROC-AUC = 0.776, PR-AUC = 0.472). The elevated AUC values across most rankings in this universe reflect the large separation between discovery genes (score > 0) and the majority of non-discovery genes (score = 0), rather than solely fine-grained within-set discrimination. For curated migraine-gene enrichment at Top-200, pathway ranking recovered 66 known genes (FE = 3.55, precision = 0.330) and hub ranking recovered 64 (FE = 3.45, precision = 0.320), both substantially exceeding the primary composite (36 genes; FE = 1.94) and significance-only (25 genes; FE = 1.35) rankings.

Within $G_{\text{discovery}}$ alone, effect-only ranking remained the strongest for test replication (ROC-AUC = 0.675, PR-AUC = 0.663), recovering 145 held-out test-positive genes at Top-200. The primary composite and significance-only rankings showed similar within-set replication (ROC-AUC $\approx$ 0.54–0.55), indicating that the headline 0.775 figure is driven in substantial part by discovery–versus–non-discovery separation rather than by within-set ranking quality alone. Pathway and hub rankings showed substantially stronger enrichment for curated migraine biology: at Top-200, pathway ranking recovered 66 known migraine genes (FE = 2.58, precision = 0.330) and hub ranking recovered 64 (FE = 2.50, precision = 0.320). Direct target evidence recovered 58 known genes (FE = 2.27, precision = 0.290), whereas drug-link count and druggability alone provided weaker standalone biological enrichment.

The integrated score provided a balanced compromise across both universes and both positive sets: it did not exceed effect-only ranking for held-out replication, nor did it exceed pathway and hub rankings for curated migraine-gene enrichment, but it retained competitive performance across all four dimensions (full-universe test ROC-AUC = 0.776; discovery-universe test ROC-AUC = 0.546; Top-200 migraine FE = 1.92, precision = 0.245). Together, these results suggest that effect size is most informative for held-out statistical replication, pathway and hub evidence better capture disease-biological coherence, and the integrated ranking provides a practical balance for downstream target selection. Full component-wise metrics across both universes are provided in Supplementary Table S4.

To assess whether the integrated score weights were arbitrary, we evaluated 17 alternative weighting schemes

spanning nine reasonable alternatives (e.g. DE-heavy, pathway-heavy, drug-heavy, equal, and no-hub variants) and eight extreme stress tests (single-component schemes). Across reasonable alternatives, the mean Spearman ρ against the default ranking was 0.963 (minimum 0.866), with a mean Top-100 gene overlap of 81.6%; crucially, performance metrics were virtually unchanged, with DISC-universe test ROC-AUC ranging only from 0.544 to 0.547 and FULL-universe test ROC-AUC from 0.775 to 0.776 across all nine schemes (Supplementary Table S1). Extreme single-component schemes diverged substantially (mean $\rho = 0.682$, minimum 0.292 for hub-only), suggesting that the integrated score draws on genuine multi-source signal rather than being dominated by any single layer. These results suggest that the default weights (DE = 0.45, Pathway = 0.25, Drug = 0.25, Hub = 0.05) lie within a stable region of parameter space, with divergence occurring only under extreme parameterizations that lack biological justification in a TWAS-first pipeline.

Significant gene comparison against Open Targets

We compared G2DR with the Open Targets migraine disease–target resource using the same curated migraine reference set ($M = 3{,}190$ genes). In the full analysis, G2DR ranked 9,305 genes across the broader genotype-first search space, whereas Open Targets returned 2,376 migraine-associated targets. The full G2DR ranking recovered 1,189 curated migraine genes (37.27%), while Open Targets recovered 1,228 (38.50%), and G2DR additionally recovered 725 curated migraine reference genes absent from Open Targets entirely. When G2DR was restricted to genes represented in the Open Targets target space ($n = 823$), Top-50 precision increased from 22.00% to 46.00% and Top-200 precision increased from 23.00% to 55.50%, approaching but not matching the standalone Open Targets precision (Top-50: 92.00%), which reflects its disease-curated, pre-filtered design rather than a direct performance comparison. Full comparison metrics are provided in Supplementary Table S6. These comparisons are complementary rather than directly competitive: Open Targets offers higher precision within a curated space, while G2DR extends target hypothesis generation beyond existing disease annotations, recovering a substantial set of curated migraine reference genes that are absent from Open Targets entirely.

Drug mapping and candidate enrichment

Using the ranked migraine gene lists (top $N = 200$ and $N = 500$), we aggregated gene–drug evidence from Open Targets, DGIdb, and ChEMBL to generate ranked candidate compounds. For $N = 200$, the pipeline compiled 5,348 gene–drug evidence rows and yielded 3,963 unique predicted drugs. For $N = 500$, evidence increased to 11,234 rows and the predicted set expanded to 7,981 unique drugs.

We evaluated overlap with a curated reference set of 4,824 normalized migraine-associated drugs under two complementary evaluation frames. First, an **ALL-drugs universe** (139,597 background drugs) tests capture and enrichment: whether the returned candidates include migraine-linked drugs above chance relative to the global pharmacopeia. Second, a **PREDICTED universe** (3,963 or 7,981 returned drugs) evaluates ranking within the returned candidate set: whether reference migraine-associated drugs are preferentially placed toward the top among returned compounds. Note that FE values below 1.0 in the PREDICTED universe reflect the fact that the reference migraine drug set exceeds the returned candidate pool in size, making standard hypergeometric enrichment interpretation inapplicable in that frame; AUROC and AUPRC are the appropriate metrics for within-set ranking performance.

Under the ALL-drugs background, both gene sets showed strong enrichment of known migraine drugs. For $N = 200$, recovery increased from 5/20 (Precision@20 = 0.25, FE = 7.23, $p = 4.93 \times 10^{-4}$) to 51/100 (Precision@100 = 0.51, FE = 14.76, $p = 4.20 \times 10^{-47}$) and 205/500 (Precision@500 = 0.41, FE = 11.86, $p = 6.19 \times 10^{-161}$). For $N = 500$, early enrichment improved further (8/20, Precision@20 = 0.40, FE = 11.58, $p = 1.75 \times 10^{-7}$; 53/100, Precision@100 = 0.53, FE = 15.34, $p = 4.48 \times 10^{-50}$; 213/500, Precision@500 = 0.426, FE = 12.33, $p = 1.67 \times 10^{-171}$). These results indicate that both gene inputs yield drug sets highly enriched for established migraine-relevant compounds relative to a global drug background.

Within the returned candidates, for $N = 200$, 353 curated migraine drugs were present among 3,963 predicted drugs and the ranking achieved AUROC = 0.8004 and AUPRC = 0.3528. For $N = 500$, 527 curated migraine drugs were present among 7,981 predicted drugs with AUROC = 0.8152 and AUPRC = 0.3311. Expanding the input from 200 to 500 genes increased the number of recovered reference drugs and modestly improved AUROC, while AUPRC decreased slightly due to the larger candidate universe. Together, the two-universe evaluation separates capture from ranking: the strong fold-enrichment and hypergeometric significance under the ALL universe supports disease-relevant signal in the upstream gene prioritization, while AUROC and AUPRC suggest that reference migraine-associated drugs tend to appear earlier than non-reference drugs within the returned set. Full multi-K overlap results for both $N = 200$ and $N = 500$ under both universes are provided in Supplementary Table S5.

When we mapped the predicted drug set back to disease indications in the aggregated drug–disease database, migraine was not always the top recovered indication. Instead, cardiometabolic, inflammatory, neuropsychiatric, and seizure-related categories frequently ranked highly. This is biologically plausible because migraine has a well-established comorbidity landscape and shares mechanisms, risk factors, and therapeutic overlap with multiple neurological and systemic disorders ([31, 32, 33, 34]). Disease-label recovery should therefore be interpreted as reflecting shared pharmacology and overlapping biology across comorbid conditions rather than as a migraine-exclusive signal. We partitioned the curated migraine drug reference set ($n = 4{,}824$ normalized drugs) into four evidence-oriented tiers: migraine-specific approved therapies (Tier 1; triptans, ditans, gepants, CGRP monoclonals, ergot derivatives), guideline-supported acute or preventive therapies (Tier 2), established off-label therapies (Tier 3), and broader literature-linked compounds (Tier 4). Recovery of each tier was evaluated within the ranked candidate drug list against a global background of 139,597 drugs.

Across tiers, no Tier 1 migraine-specific approved therapies were recovered within the Top-200 ranked drugs, with only ergotamine present anywhere in the full predicted set. This is consistent with the upstream gene prioritization: the top-ranked genes do not include the classical migraine drug targets (serotonin receptors HTR1B/HTR1D, CALCA/CALCRL) that mediate triptan and CGRP monoclonal action, so these drug classes are absent from the downstream evidence integration by design rather than pipeline failure. Tier 2 guideline-supported therapies showed modest recovery (2 drugs at Top-100; FE = 30.35, $p = 2.04 \times 10^{-3}$), including topiramate and

Table 4. Tiered migraine-evidence benchmark of the ranked drug list. Tier 1: migraine-specific approved therapies ($n_{\text{ref}} = 135$); Tier 2: guideline-supported therapies ($n_{\text{ref}} = 92$); Tier 3: established off-label therapies ($n_{\text{ref}} = 29$); Tier 4: broader literature-linked compounds ($n_{\text{ref}} = 4{,}568$). Fold enrichment (FE) and hypergeometric p-values are computed relative to a global background of 139,597 drugs. High FE values for Tiers 2–3 reflect low absolute counts and should be interpreted accordingly.

Tier	Top-20	Top-50	Top-100	Prec@100	Rec@100	FE@100	p@100	Top-200
Tier 1: Migraine-specific approved	0	0	0	0.00	0.000	0.00	1.00	0
Tier 2: Guideline-supported	0	1	2	0.02	0.022	30.35	2.04×10^{-3}	2
Tier 3: Established off-label	1	1	1	0.01	0.034	48.14	2.06×10^{-2}	1
Tier 4: Broad literature-linked	4	18	48	0.48	0.011	14.67	7.08×10^{-44}	86

amitriptyline. Tier 3 established off-label therapies showed strong fold-enrichment despite low absolute counts (1 drug at Top-100; FE = 48.14, $p = 2.06 \times 10^{-2}$), with memantine and verapamil present in the predicted set. The strongest and most statistically robust recovery was in Tier 4 broader literature-linked compounds, where 48 reference drugs were recovered at Top-100 (FE = 14.67, $p = 7.08 \times 10^{-44}$), rising to 86 at Top-200 (FE = 13.14, $p = 2.80 \times 10^{-72}$). High fold-enrichment values for Tier 2 and Tier 3 should be interpreted cautiously given small absolute overlaps; Tier 4 provides the primary statistical evidence of enrichment. Together these results indicate that the pipeline preferentially recovers broader migraine-relevant pharmacology rather than migraine-specific approved therapeutic classes, reflecting the novel and mechanism-diverse nature of the prioritized gene set (Table 4).

Directionality of prioritized gene–drug pairs

To assess mechanistic plausibility beyond drug enrichment alone, we evaluated the directionality of all gene–drug pairs derived from the top 200 genes of the final integrated migraine ranking. Across 4,861 unique gene–drug pairs, 549 (11.3%) were classified as directionally consistent, 618 (12.7%) as inconsistent, and 3,694 (76.0%) as unclear (Table 5). When restricted to the 614 pairs involving drugs from the curated migraine reference set, 50 (8.1%) were consistent, 83 (13.5%) inconsistent, and 481 (78.3%) unclear. The proportions were broadly stable across both universes, indicating that the directionality signal does not selectively concentrate in migraine-annotated compounds.

Among the 50 directionally consistent migraine-drug pairs, several biologically interpretable clusters emerged. First, aspirin–*GSTP1* (drug rank 26; *GSTP1* higher in cases; aspirin inhibits GSTP1 activity) represented a well-evidenced consistent pair supported by ChEMBL annotations and involving a known migraine gene. Second, the cardiac glycosides digoxin and digitoxin (drug ranks 126 and 133) were consistent against *ATP1A4* (higher in cases; gene rank 2), which encodes a Na^+/K^+-ATPase subunit, with inhibition of the elevated pump activity directionally compatible with the inferred signal. Third, several GLP-1 receptor agonists (exenatide, liraglutide, semaglutide, lixisenatide; drug ranks 142–146) showed consistent mappings against *ALPL* (lower in cases; gene rank 6), where agonist action is directionally compatible with reduced alkaline phosphatase expression. Fourth, α_1-adrenergic receptor antagonists including phenoxybenzamine, phentolamine, prazosin, carvedilol, and vasoactive intestinal peptide (drug ranks 377–551) were consistent against *ADRA1A* (higher in cases; gene rank 150; known migraine gene), where inhibition of elevated adrenergic signalling is mechanistically plausible. Fifth, amitriptyline (drug rank 438; a guideline-supported migraine prophylactic) showed consistent directionality against *AADAC* (higher in cases; gene rank 119) via DGIdb inhibitor annotations. A cluster of antibiotic and antiprotozoal drugs (chloramphenicol, minocycline, clarithromycin, erythromycin, and related; drug ranks 54–199) also showed consistent mappings against *MRPL36* (higher in cases; gene rank 18) through DGIdb-derived inhibitor annotations of mitochondrial ribosomal function; these pairs warrant cautious interpretation given the indirect nature of the annotation.

Directionally inconsistent mappings were observed for several top-ranked approved migraine-referenced drugs, including metformin–*NDUFV1* and metformin–*NDUFS6* (both genes lower in cases; metformin inhibits complex I, directionally opposed to the down-regulated signal), dipyridamole–*PDE4C* (lower in cases; inhibitor inconsistent with reduced expression), and multiple sodium channel blockers (carbamazepine, eslicarbazepine, oxcarbazepine, zonisamide, phenytoin, lidocaine) against *SCN5A* (lower in cases; inhibition of an already-reduced channel is directionally inconsistent). These results indicate that the framework recovers a subset of mechanistically plausible repurposing candidates concentrated around several target genes with clear disease-direction evidence, but that directionality remains unresolved for the majority of predicted pairs, reflecting limited mechanism-of-action annotation coverage in current drug–target databases.

Top gene and drug validation panel

To provide a concise summary of the final integrated prioritization, we assembled top-gene and top-drug evidence panels from the Combined_Score-ranked gene list and the downstream drug-mapping and directionality outputs. The top-gene panel (Table 7) reports the 10 highest-ranked genes together with their evidence support, inferred disease direction, migraine relevance, and linked druggability information. The top-drug panel (Table 8) reports the 10 highest-ranked drugs with merged target gene information, approval status, and directionality classification.

Top gene highlights.

Among the top 10 integrated-score genes, *ATP1A4* (rank 2; higher in cases; known migraine gene; 24 linked drugs; Consistent directionality) was the most directly disease-relevant, encoding the Na^+/K^+-ATPase $\alpha 4$ subunit and linking to cardiac glycoside inhibitors including digoxin and digitoxin. *ALPL* (rank 6; lower in cases; Tier 1 confidence; 96 linked drugs; Consistent) was the most druggable top gene, linked to GLP-1 agonists through DGIdb annotations. *HLA-DQB1* (rank 3; lower in cases; known migraine gene; 55 linked drugs) had the broadest drug connectivity but showed inconsistent directionality. *MRPL36* (rank 18; higher in cases; 67 linked drugs; Consistent) was linked to a cluster of antibiotic and antimicrobial drugs through mitochondrial ribosomal inhibition annotations. Among the 49 known migraine genes present in the top 200, additional highly ranked examples included *GSTP1* (rank 31; Consistent; aspirin-GSTP1 was the most evidenced

Table 5. Summary of directionality classification for unique prioritized gene–drug pairs from the final integrated migraine ranking. Results are shown for all pairs derived from the top 200 ranked genes and separately for pairs involving drugs from the curated migraine reference set ($n_{\text{mig}} = 614$).

Directionality class	All pairs ($n = 4{,}861$)		Migraine drugs only ($n = 614$)	
	Count	%	Count	%
Consistent	549	11.3%	50	8.1%
Inconsistent	618	12.7%	83	13.5%
Unclear	3,694	76.0%	481	78.3%
Total	4,861	100.0%	614	100.0%

Table 6. Representative directionally consistent gene–drug pairs from the migraine reference set. Pairs were classified as consistent when the known drug action was mechanistically compatible with the inferred disease-associated direction of the target gene. Drug rank is from the integrated gene-to-drug scoring pipeline; gene rank is from the Combined_Score integrated gene prioritization. Evidence source indicates the database providing mechanism-of-action annotation. The *MRPL36* antibiotic cluster (rows 3–9) reflects DGIdb-derived mitochondrial ribosomal inhibitor annotations and should be interpreted with caution.

Drug rank	Drug	Gene	Gene rank	Disease direction	Drug action	Evidence source	Approved	Known migraine gene
26	Aspirin	GSTP1	31	Higher in cases	Inhibitor	ChEMBL	Yes	Yes
126	Digoxin	ATP1A4	2	Higher in cases	Inhibitor	DGIdb	Yes	Yes
133	Digitoxin	ATP1A4	2	Higher in cases	Inhibitor	DGIdb	Yes	Yes
142	Exenatide	ALPL	6	Lower in cases	Agonist	DGIdb	Yes	No
144	Liraglutide	ALPL	6	Lower in cases	Agonist	DGIdb	Yes	No
146	Semaglutide	ALPL	6	Lower in cases	Agonist	DGIdb	Yes	No
54	Chloramphenicol	MRPL36	18	Higher in cases	Inhibitor	DGIdb	Yes	No
86	Minocycline	MRPL36	18	Higher in cases	Inhibitor	DGIdb	Yes	No
162	Clarithromycin	MRPL36	18	Higher in cases	Inhibitor	DGIdb	Yes	No
379	Prazosin	ADRA1A	150	Higher in cases	Inhibitor	DGIdb	Yes	Yes
512	Carvedilol	ADRA1A	150	Higher in cases	Inhibitor	DGIdb	Yes	Yes
438	Amitriptyline	AADAC	119	Higher in cases	Inhibitor	DGIdb	Yes	No
257	Empagliflozin	SLC5A2	45	Higher in cases	Inhibitor	DGIdb	Yes	No
222	Glycine	GRIN2B	44	Lower in cases	Agonist	DGIdb	Yes	Yes
497	Loxapine	KCNT1	133	Lower in cases	Activator	DGIdb	Yes	Yes

approved consistent pair), *GRIN2B* (rank 44; 110 linked drugs; known migraine gene), and *OSGEP* (rank 58; highest composite Importance_Score of 0.682; known migraine gene). Several top-ranked genes — including *GRHPR*, *MRPL21*, *DHX15*, and *GCDH* — had few or no linked drugs, indicating prioritized signals not yet accessible to repurposing through current drug–target databases.

Top drug highlights.

Among the top 10 ranked drugs, lifitegrast (rank 6; approved; targets *ITGB2* and *ICAM1*, both higher in cases; Consistent) was the only approved drug in the top 10 with directional consistency, acting as an integrin inhibitor compatible with elevated integrin subunit expression. Oleclumab (rank 12; Phase 3; targets *NT5E*; Consistent) and ezatiostat (rank 13; Phase 2; targets *GSTP1*; Consistent) provided the clearest non-approved consistent candidates. Neramexane (rank 17; Phase 3; targets *GRIN2B* and *CHRNA10*; Consistent) was the highest-ranking investigational drug with a migraine-biology rationale through NMDA receptor modulation. Several top-ranked approved drugs showed directionally inconsistent mappings, including metformin (rank 2; Inconsistent against *NDUFV1* and *NDUFS6*), amantadine (rank 5; Inconsistent against *GRIN2B*), dibotermin alfa (rank 4; Inconsistent against *BMPR1B*), and apremilast (rank 10; Inconsistent against *PDE4C*). Among the top 20 migraine-reference drugs, aspirin (rank 26; Consistent against *GSTP1*) was the highest-ranking approved drug with directional support, while metformin, esketamine, and memantine appeared at ranks 2, 7, and 8 respectively, consistent with their presence in the migraine drug compendium.

Biological interpretation of prioritised genes and drug repurposing candidates

The 200 genes prioritised by the integrated G2DR pipeline converged on several biological themes with established or plausible relevance to migraine pathophysiology. Of these, 49 (24.5%) had prior evidence linking them to migraine or closely related headache phenotypes based on database annotation from OMIM, DisGeNET, and HPO, whereas the remainder represent computationally prioritized candidates without established migraine association. These novel signals should be treated as hypothesis-generating outputs of the prioritization pipeline rather than validated migraine targets, and their biological interpretation below is offered to motivate experimental follow-up rather than to make mechanistic claims. Overall, the prioritised genes mapped to pathways related to glutamatergic and ion-channel excitability, mitochondrial and metabolic function, immune and inflammatory signalling, cerebrovascular and extracellular-matrix integrity, and broader neuronal and synaptic maintenance, consistent with the multifactorial biology highlighted by large migraine genetic studies [38, 39].

Among the most interpretable signals, *GRIN2B* and *SLC1A1* supported glutamatergic hyperexcitability and cortical-spreading-depression-related biology [40, 41], while *SCN1B*, *KCNT1*, *KCNS2*, and *ATP1A4* pointed to broader disturbance of ion-channel homeostasis and neuronal firing. A second major component comprised mitochondrial and metabolic genes, including *NDUFV1*, *NDUFS6*, *MRPL21*, *MRPL36*, *TUFM*, *VARS2*, *FXN*, *ETFB*, *COASY*, and *HAAO*, consistent with longstanding hypotheses that impaired cerebral bioenergetics contributes to migraine susceptibility [42, 43]. Immune and inflammatory signals were also prominent, including multiple

Table 7. Top 10 prioritized genes from the final integrated migraine ranking. Genes are ranked by Combined_Score (integrated: DE + pathway + druggability + hub). KnMig = known migraine gene; Tier = confidence tier from primary composite ranking; NDrugs = number of unique linked drugs; ApprDrug = has at least one approved linked drug; DirBest = best directionality support across linked drug pairs.

Rank	Symbol	CombScore	ImpScore	Disease direction	KnMig	Confidence tier	Hits	Tissues	DBs	NDrugs	ApprDrug	DirBest
1	APOBEC3G	0.936	0.435	Higher in cases	No	Tier3_Exploratory	100	29	1	98	Yes	Unclear
2	ATP1A4	0.934	0.531	Higher in cases	Yes	Tier3_Exploratory	18	3	3	24	Yes	Consistent
3	HLA-DQB1	0.933	0.410	Lower in cases	Yes	Tier2_Moderate	32	15	2	55	Yes	Inconsistent
4	PSMB5	0.932	0.413	Lower in cases	No	Tier2_Moderate	40	13	2	33	Yes	Inconsistent
5	SLC1A1	0.932	0.491	Lower in cases	No	Tier2_Moderate	103	15	2	54	Yes	Inconsistent
6	ALPL	0.931	0.412	Lower in cases	No	Tier1_High	35	5	1	96	Yes	Consistent
7	AQP5	0.930	0.419	Lower in cases	No	Tier2_Moderate	42	10	2	29	Yes	Consistent
8	CBL	0.930	0.385	Unclear	No	Tier3_Exploratory	4	2	1	29	Yes	Unclear
9	GRHPR	0.929	0.485	Higher in cases	No	Tier3_Exploratory	206	16	1	2	No	Unclear
10	MRPL21	0.929	0.447	Higher in cases	Yes	Tier2_Moderate	93	31	1	1	No	Unclear

Table 8. Top 10 prioritized drug candidates from the final integrated migraine repurposing analysis. Drugs are ranked by DrugScore; target genes are merged per drug. KnMig = drug present in curated migraine reference set; NTarg = number of unique target genes among top 200; Dir = best directionality classification across gene–drug pairs.

Rank	Drug	Phase	KnMig	Approved	NTarg	Target genes	Disease directions	Directionality	Evidence sources
1	Ataluren	APPROVED	No	Yes	4	RPL28, RPS14, RPL12, RPS16	Unclear, lower	Unclear	DGIdb, OpenTargets
2	Metformin	APPROVED	Yes	Yes	2	NDUFV1, NDUFS6	Lower in cases	Inconsistent	OpenTargets, DGIdb
3	Voxelotor	APPROVED	No	Yes	1	HBB	Lower in cases	Unclear	OpenTargets
4	Dibotermin alfa	APPROVED	No	Yes	1	BMPR1B	Higher in cases	Inconsistent	OpenTargets
5	Amantadine	APPROVED	No	Yes	1	GRIN2B	Lower in cases	Inconsistent	OpenTargets
6	Lifitegrast	APPROVED	No	Yes	2	ITGB2, ICAM1	Higher in cases	Consistent	OpenTargets
7	Esketamine	APPROVED	Yes	Yes	1	GRIN2B	Lower in cases	Unclear	OpenTargets, DGIdb
8	Memantine	APPROVED	Yes	Yes	1	GRIN2B	Lower in cases	Unclear	OpenTargets, DGIdb
9	Midostaurin	APPROVED	No	Yes	7	CBL, BMPR1B, CDK3, LATS2, PRKD3, SEPTIN9, PEG3	Mixed	Inconsistent	OpenTargets, DGIdb
10	Apremilast	APPROVED	No	Yes	1	PDE4C	Lower in cases	Inconsistent	OpenTargets

HLA-region genes together with *RELA*, *ICAM1*, *CLEC7A*, *LY96*, *UNC93B1*, and *SERPINE1*, in keeping with evidence that neuroinflammatory and immune-trafficking mechanisms contribute to trigeminovascular sensitisation [44, 45, 46]. In parallel, neurovascular and extracellular-matrix genes such as *HTRA1*, *HSPG2*, *GNAQ*, *TIE1*, *EGFL7*, *LAMB1*, *F12*, and multiple collagen-related genes supported a vascular-integrity component, consistent with the broader overlap between migraine and cerebral small-vessel or neurovascular disorders [47, 48, 49]. Together, these results suggest that the prioritised set reflects convergence across excitability, metabolism, inflammation, and neurovascular maintenance rather than a single dominant molecular pathway.

The downstream drug-mapping analysis likewise recovered a broad migraine-relevant pharmacological landscape rather than only migraine-specific therapies. Using the top 200 genes, the pipeline identified 3,963 unique candidate drugs from 5,348 gene–drug evidence rows, and 353 curated migraine-associated drugs were present within the predicted set. Enrichment against the global drug background was strong, indicating that the upstream gene prioritisation captures disease-relevant pharmacology even though the output did not capture classical migraine medications. At the top of the ranked list, several recovered migraine-linked drugs were already notable, including metformin (rank 2), esketamine (rank 7), memantine (rank 8), aspirin (rank 26), dexamethasone (rank 48), lidocaine (rank 66), zonisamide (rank 79), indomethacin (rank 85), and acetaminophen (rank 98). Additional recovered migraine-associated or migraine-relevant drugs were present lower in the ranked set, including verapamil (rank 213), valproic acid (rank 253), celecoxib (rank 340), diclofenac (rank 343), amitriptyline (rank 438), topiramate (rank 491), and ergotamine tartrate (rank 521), showing that the pipeline recovered both acute-care and preventive classes as well as broader off-label and mechanism-linked therapies [50, 51, 52, 53, 54, 55, 56, 57].

Directionality filtering was particularly useful for separating broadly recovered drugs from those with clearer mechanistic compatibility. Among migraine-linked drugs, aspirin showed consistent directionality at *GSTP1*; amitriptyline was directionally consistent at *AADAC*; and a cluster of antibiotic-derived mappings including chloramphenicol, minocycline, clarithromycin, erythromycin, tigecycline, and tobramycin were directionally consistent through *MRPL36*, although these latter signals should be interpreted cautiously because they arise from mitochondrial ribosomal annotations rather than direct migraine therapeutic use. Several non-reference but biologically interesting candidates were also directionally consistent as computational hypotheses, including lifitegrast at the *ITGB2–ICAM1* axis, neramexane at *GRIN2B/CHRNA10*, digoxin and digitoxin at *ATP1A4*, and GLP-1 receptor agonists such as exenatide, liraglutide, and semaglutide at *ALPL*; the latter is of potential interest given recent emerging clinical reports of GLP-1 pathway modulation in chronic migraine [58, 59, 60, 61], although all such pairs require direct experimental and clinical validation before any repurposing conclusions can

be drawn. By contrast, several high-ranking recovered migraine drugs were directionally inconsistent, including metformin at *NDUFV1/NDUFS6*, dipyridamole at *PDE4C*, carbamazepine, zonisamide, eslicarbazepine, oxcarbazepine, phenytoin, and lidocaine at *SCN5A*, as well as verapamil in its mapped targets. Other prominent recovered drugs, including esketamine, memantine, dexamethasone, indomethacin, acetaminophen, and valproic acid, were retained but remained directionally unresolved. Overall, these results indicate that the G2DR framework recovers a substantial set of migraine-relevant compounds, but that directional filtering is necessary to distinguish broadly associated drugs from those with stronger mechanistic coherence for repurposing.

Discussion

We present G2DR as a genotype-first computational framework for genetics-informed target prioritization and drug repurposing hypothesis generation in settings where genotype and phenotype labels are available but matched disease transcriptomics are limited or absent. The principal contribution is not a new TWAS algorithm or a definitive therapeutic predictor, but an integrative workflow that converts inherited genetic signal into ranked target and compound hypotheses through multiple complementary evidence layers. Critically, G2DR is designed as a prioritization engine that narrows the candidate space for experimental and translational follow-up, not as a system for identifying clinically actionable targets or providing ready-to-use repurposing recommendations; all prioritized genes and drug candidates require independent pharmacological, mechanistic, and clinical validation before any translational conclusions can be drawn. In a migraine case study, the framework generalized to held-out data at the gene-prioritization stage, recovered established migraine biology, and generated drug candidate sets enriched for known migraine-relevant therapies relative to a global drug background. These findings support the utility of genetics-anchored, modular integration for computational hypothesis generation in genotype-first settings. Several aspects of the results are particularly informative. First, transcriptome-prediction resources were only moderately concordant, reinforcing the importance of evaluating multiple expression-weight models rather than assuming a single reference architecture is sufficient. Second, pathway and network information contributed substantially to biological prioritization, suggesting that translationally useful targets are more effectively recovered when statistical association is interpreted in broader mechanistic context. Third, drug mapping recovered many clinically familiar and biologically plausible migraine-linked compounds, especially among off-label and shared-mechanism therapeutic classes. Together, these results argue that the strongest use case for G2DR is structured target prioritization and repurposing hypothesis generation rather than direct inference of treatment efficacy. Component-level benchmarking (Supplementary Table S4) and contextual comparison with Open Targets (Supplementary Table S6) provide internal reference points, but a formal head-to-head evaluation against published genetics-to-drug prioritization pipelines remains an important direction for future work. To assess causal support for prioritized genes, we performed single-instrument Mendelian randomization using GTEx v8 cis-eQTLs and large-scale migraine GWAS summary statistics across ranked gene sets at multiple depth thresholds. While nominally significant associations emerged in broader ranked sets, no gene survived multiple-testing correction at FDR < 0.10, and colocalization identified no gene–locus pairs with PP.H4 ≥ 0.80. These null findings most likely reflect limited cis-eQTL instrument availability and the modest statistical power of the migraine cohort, rather than an absence of true genetic signal in the prioritized genes. Across cross-validation folds, gene-level rankings showed limited exact overlap at the gene-identity level (mean pairwise Jaccard: 0.019 at Top–50; 0.045 at Top–200). This is expected in small, class-imbalanced cohorts where each fold contains only approximately ∼42–43 cases in training and ∼5–6 cases in validation, making individual gene rankings susceptible to sampling variation. Critically, however, fold-wise prioritization performance remained stable (mean ROC-AUC = 0.675, SD = 0.004; mean PR-AUC = 0.149, SD = 0.004), indicating that the ranking signal—the ability to separate true-positive genes from negatives—is reproducible even when the specific membership of top-ranked gene lists varies across folds. This distinction between gene-identity overlap and ranking-performance stability is important: the latter is the relevant metric for a prioritization framework, and it was consistent across folds. Drug-level rankings were substantially more stable (mean pairwise Jaccard: 0.475 at Top-20; 0.688 at Top-50), suggesting that the downstream repurposing signal is robust to fold composition. The current study has important limitations. The migraine cohort is modest and class-imbalanced, and the framework relies on genetically predicted rather than measured context-specific transcription. Prioritized genes should therefore be interpreted as genetically supported candidates rather than confirmed causal effector genes. TWAS-style signal can be confounded by linkage disequilibrium and co-regulation, and drug mapping does not by itself resolve causal directionality, target modulation direction, or clinical viability. These limitations position G2DR explicitly as a computational prioritization engine for downstream experimental and clinical validation, not as a stand-alone therapeutic decision system or a source of clinically actionable target recommendations. Users should treat all prioritized genes and drug candidates as computational hypotheses: they require orthogonal validation including functional studies, independent cohort replication, and where feasible, controlled experimental perturbation before any translational value can be claimed. The drug-level results are also informative in what they do not yet capture well. The framework preferentially recovered broad off-label and mechanism-linked migraine therapies, whereas modern migraine-specific classes — including serotonergic and CGRP-axis agents — were comparatively under-represented. This likely reflects both database curation bias and the tendency of genetically prioritized gene programs to map more readily onto shared neurobiological processes than onto highly disease-specific pharmacological axes. Future versions of G2DR should incorporate stronger causal filters such as fine-mapping and colocalization, as well as directionality-aware resources including perturbational signatures and mechanism-of-action annotations. Overall, G2DR provides a useful foundation for genetics-informed repurposing in genotype-first settings. Its modular design allows independent refinement of transcriptome imputation, gene-level testing, biological contextualization, and drug mapping, making it adaptable to additional traits and larger cohorts. The next critical step is not simply scaling the pipeline, but strengthening causal attribution and translational specificity through external validation and more stringent benchmarking. Framed in this way, G2DR is a proof-of-concept

framework for target and drug prioritization with clear room for further maturation.

Key Points

- G2DR is a genotype-first computational prioritization engine for hypothesis generation in settings where matched disease transcriptomics are unavailable; it is designed to inform downstream follow-up rather than to provide clinically actionable target recommendations directly.
- The framework integrates genetically predicted gene expression across seven transcriptome-weight resources, multi-method gene-level testing, pathway enrichment, network context, and multi-source drug–target evidence into a modular and reproducible prioritization pipeline.
- In internal held-out evaluation, gene prioritization generalized to test data (ROC-AUC = 0.775; PR-AUC = 0.475), with fold enrichment > 2.9-fold for test-positive genes even at Top-2,000, supporting discovery without test leakage.
- Drug mapping from 200 prioritized genes yielded 3,963 candidate compounds with 14.76-fold enrichment for migraine-associated drugs at Top-100 relative to a global background of 139,597 drugs; directionality filtering identified mechanistically coherent candidates including lifitegrast, neramexane, GLP-1 agonists, and α_1-adrenergic blockers as computational hypotheses for experimental follow-up.
- All prioritized genes and drug candidates are computational hypotheses requiring independent experimental, pharmacological, and clinical validation; G2DR is a proof-of-concept framework demonstrated in migraine with clear directions for future improvement in causal attribution and translational specificity.

Competing interests

The authors declare no competing interests.

Author contributions statement

M.M. conceived the study, designed and implemented the G2DR framework, performed all analyses, and drafted the manuscript. D.B.A. supervised the project, contributed to study design and interpretation, and reviewed and edited the manuscript. All authors approved the final manuscript.

Funding

This work was supported by the University of Queensland and the Baker Heart and Diabetes Institute. The UK Biobank data were accessed under application ID 50000. The funding sources had no role in study design, data collection, analysis, interpretation, or the decision to submit for publication.

Data and Software Availability

The genotype and phenotype data used in this study were accessed through the UK Biobank under application ID 50000 (`https://www.ukbiobank.ac.uk/`) and are subject to UK Biobank access restrictions; researchers may apply for access through the UK Biobank Access Management System. The G2DR framework source code is freely available for non-commercial use at `https://github.com/MuhammadMuneeb007/G2DR-A-Genotype-First-Framework-for-Genetics-Informed-Target-Prioritization-and-Drug-Repurposing`. Supporting utilities are available at the following repositories: gene phenotype downloader, `PhenotypeToGeneDownloaderR` (`https://github.com/MuhammadMuneeb007/PhenotypeToGeneDownloaderR`); gene identifier converter, `GeneMapKit` (`https://github.com/MuhammadMuneeb007/GeneMapKit`); drug–disease downloader, `DownloadDrugsRelatedToDiseases` (`https://github.com/MuhammadMuneeb007/DownloadDrugsRelatedToDiseases`). All repositories will be maintained for a minimum of two years following publication. Public annotation resources used by the pipeline include STRING (`https://string-db.org`), Open Targets (`https://www.opentargets.org`), DGIdb (`https://www.dgidb.org`), and ChEMBL (`https://www.ebi.ac.uk/chembl`).

Acknowledgments

Not applicable

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